PRODUCT REQUIREMENTS DOCUMENT (PRD)

MemoryVault AI

Context-Aware Misplaced Item Prediction & Search Route Optimization System

Document Status: Approved for MVP **Target Stack:** Python, Flask, scikit-learn **Dataset:** ~5,000 Synthetic Records

1. Problem Statement

People frequently misplace everyday, untagged items (such as keys, wallets, spectacles, and earphones) and rely on flawed human memory recall, especially during moments of distraction, stress, or departures from usual routines. Existing market solutions fall into two deficient extremes: physical hardware trackers (e.g., Bluetooth tags), which are expensive, bulky, and cannot be affixed to every item, or passive note-taking apps, which require unnatural manual logging. There is currently no software-only intelligence layer that reasons about contextual behavioral patterns and daily habits to probabilistically predict item locations.

2. Target Users

- **Busy Professionals & Students:** Individuals subject to frequent routine shifts and high cognitive loads leading to absent-minded placements.
- **Neurodivergent Individuals:** Individuals diagnosed with ADHD or executive function challenges who experience object permanence blindspots.
- **Elderly Users:** Individuals seeking lightweight, conversational cognitive support for routine home management.
- **Minimalists:** Users seeking the utility of item tracking without purchasing, charging, or adhering physical hardware tags.

3. Goals & Objectives

- **Probabilistic Spatial Prediction:** Accurately infer the top candidate locations using situational telemetry (temporal, weather, and activity context).
- **Physical Search Optimization:** Synthesize probability scores with spatial home topology into an ordered, friction-minimized check sequence.
- **Frictionless Interaction:** Provide an intuitive conversational chatbot interface that minimizes conversational overhead and replaces static forms.

4. Core Technical Architecture & Features

MemoryVault AI addresses the full academic and practical machine learning lifecycle across three integrated pillars:

Component	Methodology / Pipeline	Functional Role
Contextual Habit Profiling	Unsupervised Learning: K-Means Clustering + Principal Component Analysis (PCA)	Discovers latent behavioral archetypes (e.g., "Late-night fatigued entry" vs. "Midday grocery arrival") across multi-dimensional feature sets.
Location Prediction Engine	Supervised Learning: Multiclass Random Forest Classifier	Maps contextual vectors (time, weather, day, current activity) to ranked probability distributions across indoor candidate zones.
Smart Search Router	Search Algorithm: A* / Greedy Best-First Graph Search	Calculates the optimal physical search sequence by balancing candidate location probabilities against inter-room walking distances.

Component	Methodology / Pipeline	Functional Role
Chatbot & Serving Layer	Flask REST API + Lightweight Web UI	Conducts conversational Q&A to resolve contextual uncertainty and displays the final ranked search plan.

5. MVP Scope & Technical Specifications

The Minimum Viable Product focuses on a single-user residential environment modeled as a standard topological room graph.

- **Synthetic Dataset Generation:** Synthetic generation of ~5,000 records capturing timestamps, rain/temperature status, entry points, carried items, and ground-truth recovery coordinates.
- **Model Evaluation Protocol:** Strict validation regimen including 5-fold Stratified Cross-Validation, Confusion Matrix generation, and multi-class One-vs-Rest ROC-AUC evaluation.
- **Graph Topology Engine:** Standard indoor layout represented as an adjacency matrix with assigned traversal costs (distance/time) for A* pathing.

6. User Stories

- **US1:** As a user, I want to ask the chatbot, *"Where did I leave my keys?"* so I can immediately receive likely locations without needing pre-installed hardware.
- **US2:** As a user, I want the bot to ask 1–2 quick clarifying context questions (e.g., *"Were you carrying groceries when you returned?"*) to dramatically sharpen prediction accuracy.
- **US3:** As a user, I want an ordered search checklist (e.g., 1. Kitchen counter → 2. Hallway credenza → 3. Sofa cushion) so that I do not redundantly pace back and forth.
- **US4:** As a system evaluator, I want model performance visualizations (confusion matrices, ROC curves) accessible via endpoints to evaluate prediction efficacy.

7. Success Metrics & Key Performance Indicators (KPIs)

Metric Category	Target KPI	Validation Criteria
Top-3 Categorical Accuracy	≥ 80.0%	Target item exists within top 3 predicted zones across holdout test sets.
Discrimination Capability	ROC-AUC ≥ 0.85	Macro-averaged One-vs-Rest ROC-AUC across all domestic zones.
Inference & Route Latency	< 2.0 seconds	End-to-end turnaround: chatbot prompt input to rendered search checklist.
Search Path Efficiency	≥ 95.0% Superiority	A* traversal cost lower than randomized search order permutations.

8. Assumptions & Dependencies

- Synthetic distributions (~5,000 records) sufficiently approximate real-world domestic habit entropy for baseline classification.
- Indoor residential zones can be abstracted into discrete graph nodes (e.g., Foyer, Kitchen Island, Desk, Nightstand).
- Users accept answering up to two conversational clarifying prompts if search latency and precision are optimized.

9. Risks & Mitigations

- **Cold-Start Anomaly:** New users lack personalized historical logs.
  Mitigation: Pre-train model weights on the generalized synthetic corpus; use transfer fine-tuning as real feedback logs accumulate.
- **Pathological Search Routing:** A* prioritizing a low-cost nearby room with negligible item probability.
  Mitigation: Construct cost function balancing edge distance inversely against zone likelihood: $Cost = Distance / (P_{location} + \varepsilon)$.
- **Class Imbalance in Infrequent Zones:** Rare placement locations suffer from low recall.
  Mitigation: Employ SMOTE or stratified sampling during synthetic dataset generation and cross-validation splitting.

10. Out-of-Scope Features (Strictly Excluded from MVP)

- Native mobile applications (iOS/Android) or widget support.
- Hardware integrations with Apple FindMy, BLE beacons, Tile, or computer vision cameras.
- Multi-building or transit tracking (e.g., car interiors, office cubicles, public transit).
- Real-time speech-to-text / voice interfaces (interactions are strictly text-based).

11. Acceptance Criteria

- **Supervised Model:** scikit-learn Random Forest classifier trained, outputting calibrated probability distributions alongside evaluated Confusion Matrix and ROC-AUC curves.
- **Unsupervised Pipeline:** PCA applied for dimensionality reduction; K-Means producing identifiable, distinct habit clusters.
- **A* Routing Engine:** Graph search module accepting probabilistic output and returning an ordered, cost-minimized physical path.
- **Interactive Delivery:** Flask server running locally, rendering a functional chatbot interface processing inquiries and returning search sequences.